UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

Akesh

END SEMESTER EXAMINATION
SEMESTER II - 2014/2015

General Physics II - PH 102

Time allowed: *Two hours*

Answer **FIVE** questions selecting **TWO** questions from Section A, **TWO** questions from Section B and **ONE** question from Section C.

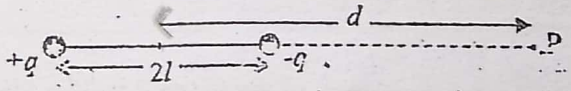
(Ccalculators are allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_o)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}.

(Submit answers for questions in Section A separately)

SECTION – A

1. (a) Clearly identifying the quantities involved, write down Coulomb's law in vector form.



(b) An electric dipole has charges $+q$ and $-q$ separated by a distance $2l$ as shown in the figure. Point P is situated on the axis at a distance d from the mid point of the dipole.

(i) Obtain an expression for the electric field intensity at point P on the axis.

(ii) Also obtain an expression for the electric potential at point P.

(c) If the dipole is now rotated in clockwise direction about the mid point through an angle 90°, find the new value of the electric field intensity and the electric potential at point P.

2. (a) With the aid of a clearly labeled diagram, explain the action of a cyclotron.

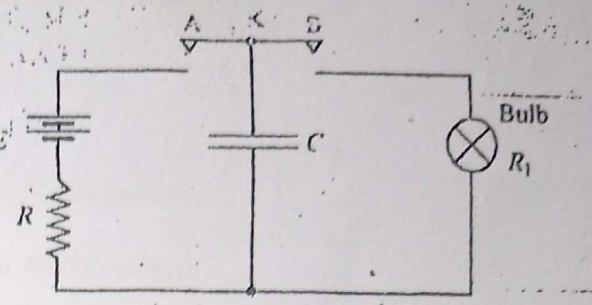
(b) A cyclotron has dees of radius R and a magnetic field of flux density B. Obtain an expression for the cyclotron frequency that is needed to accelerate ions of charge q and mass m.

(c) If $R = 50$ cm and $B = 2.0$ T, find the cyclotron frequency necessary to accelerate alpha particles of mass 6.65×10^{-27} kg and charge 3.2×10^{-19} C.

(d) What is the maximum kinetic energy of alpha particles coming out from the above cyclotron?

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3. The flash circuit of a camera is shown in the figure. The e.m.f. of the battery is E and it has a negligible internal resistance. R is the resistance of a resistor in the circuit and R_1 is the resistance of the bulb. The capacitor C is charged by pressing the end A of the two-way key K at time $t = 0$.



(a) Suppose that the end A is kept pressed for a period of time t.

(i) Obtain an expression for the charge q accumulated in the capacitor C at time t.

(ii) Give the time constant of this part of the circuit.

(iii) Also write down an expression for the current I through the circuit at time t.

(iv) If the e.m.f of the cell $E = 6.0$ V, the value of $C = 2.0$ mF and the value of $R = 50\ \Omega$, find the value of the time constant τ and the charge at $t = 5\tau$.

(c) After charging the capacitor to its maximum theoretical value, the end B of the key is pressed starting $t = 0$ to discharge the capacitor through R_1. Without detailed proof, write down an expression for the current in the second circuit at time t. Find the value of the resistance R_1 in order to get an initial current of 1.0 A through the bulb.

(Submit answers for questions in Section B and C separately)

SECTION B

4. (a) A photon of wavelength 0.0003 nm in the vicinity of a heavy nucleus produces an electron positron pair. If the kinetic energy of the positron is twice that of the electron, determine the kinetic energy of each of the particles.
[Rest energy of an electron = 0.511 MeV. Planck constant = 4.136×10^{-21} MeV s. Speed of light in free space = 3.00×10^8 m s^{-1}.]

(b) A monoenergetic beam of electrons is incident on a single slit of width 0.5 nm. A diffraction pattern is observed on a screen 20 cm from the slit. The half width of the central maximum of the diffraction pattern is 2.1 cm. [Angular half width of the central maximum of a diffraction pattern is given by $\sin\theta = \frac{\lambda}{a}$, where a is the width of the single slit and λ is the wavelength of the light used.] Planck's constant = 6.63×10^{-34} J s.

(i) Determine the wavelength corresponding to the diffraction pattern.

(ii) Find the de Broglie wavelength of an electron in the beam.

(iii) Hence find the momentum of an electron in the beam.

5. (a) According to the Bohr model, the energy levels of the electron of the hydrogen atom are given by

$$E_n = (-13.6\ \text{eV})\frac{1}{n^2}.$$

(i) Find the shortest and the longest wavelengths of the Lyman series of hydrogen spectrum. [Planck constant = 4.136×10^{-15} eV s and speed of light in free space = 3.00×10^8 m s^{-1}.]

(ii) Using your results in part (i) find the Rydberg constant (R).

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(b) (i) Find the activity of one gram of $^{226}_{88}Ra$ in Ci. The half life of $^{226}_{88}Ra$ is 1622 years. [1 Ci = 3.7×10^{10} Bq.]

(ii) Also find the activity of the sample 1000 years later.

6. (a) When $^{6}_{3}Li$ is bombarded with 4 MeV deuterons (nucleus of deuterium $^{2}_{1}H$), one reaction that is observed is the formation of two alpha particles, each with 13.2 MeV of energy

(i) Write the equation for the above nuclear reaction.

(ii) Find the Q value for the reaction.

(iii) Find the mass of $^{6}_{3}Li$. The masses of He and deuterium are 3727.4 MeV/c^2 and 1875.6 MeV/c^2 respectively.

(b) Find the binding energy of deuteron (nucleus of deuterium $^{2}_{1}H$). The atomic masses of deuterium and hydrogen are 2.014102 u and 1.007825 u respectively and the mass of neutron is 1.008665 u. 1 u = 931.5 MeV/c^2.

(c) (i) Name the four fundamental forces and their corresponding mediating field particles.

(ii) The charges of the u, d and s quarks are $+\frac{2}{3}e$, $-\frac{1}{3}e$ and $-\frac{1}{3}e$ respectively. If the quark composition of the proton is uud determine the quark composition of the neutron. If the quark composition of π^+ is $u\bar{d}$ determine the quark composition of π^-.

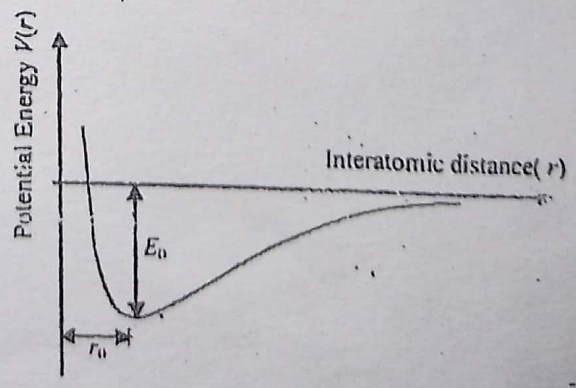
SECTION – C

7. (a) Explain the three bond types (1) ionic, (2) covalent and (3) metallic.

(b) What are polar molecules?

(c) Explain the Van der Waals force and the Van der Waals bonds.

8. The figure shows the potential energy function of two ions as a function of their separation r. The approximate expression for the potential energy is given by $V(r) = -\frac{ke^2}{r} + \frac{b}{r^9}$. The first term is due to the usual Coulomb interaction and the other term is introduced to account for the repulsive effects of the two ions at small distances.



(a) Determine b as a function of the equilibrium spacing r_0.

(b) Determine an expression for the minimum potential energy E_0 as a function of the equilibrium spacing r_0.

(c) For KCL (potassium chloride) $r_0 = 0.279$ nm. Calculate E_0 for KCL. [$ke^2 = 144.0$ eV nm.]

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